

Bridge Day 8 STUDENT QUIZ

Debugging, Boundary Tests, and Complete Repair Evidence

Learning sequence: Predict - Trace - Explain - Test - Interpret

Wednesday, July 15, 2026 • 20 points • target 17 minutes

Name		UIN		Section	
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Assessment boundary and evidence source

Contract: 07a04826c975

Cumulative foundations: input conversion, comparisons, Boolean repair, expected vs actual, boundary tests

Scaffold level: complete program plus protective test set

Evidence source: the matching v5 workbook readiness/evidence pages and VIP activities 18.13, 18.14, 18.15, 18.16, 18.17.

Show intermediate state for trace credit. Program credit requires one coherent solution, a stated assumption when the prompt leaves one implicit, and at least one test whose expected result can distinguish correct from incorrect logic.

Item 1. Diagnose an inclusive-boundary defect. - 5 points

```
temperature = 35
ready = temperature > 15 and temperature < 35
print(ready)
```

Question: The requirement includes 15 and 35. Give actual result, cause, corrected assignment, and two tests that expose the defect.

Item 2. Build a minimum protective boundary set. - 5 points

```
reading = 10
low = 10
high = 20

if reading < low:
    status = "below"
elif reading > high:
    status = "above"
else:
    status = "within"

print(status)
```

Question: Give exact supplied output, then specify four tests that protect both inclusive boundaries and their immediate outside cases.

Complete program - 10 points

- Read reading, low, and high as floats.
- Print invalid when low is greater than high.
- Otherwise print Result: below, Result: within, or Result: above. Equality at either limit is within.
- Use one mutually exclusive chain and provide four boundary tests with expected output.

Program workspace**Test input, expected result, and why this test matters**